Introduction to Data Science

Department of FST

Final Term Project

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Section: [E]

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Project Title: Cross-Domain Knowledge Mapping: Biomedical vs AI Research.

Introduction: This project is based on rapid growth of research in both biomedical science and artificial intelligence (AI). It is important to understand how these domains are related. This project focuses on analyzing research abstracts from both domains using data science techniques to discover patterns, similarities, and differences between biomedical and AI research using text mining, dimensionality reduction, and clustering methods using R.

Dataset & Libraries:

```
if (!require("pacman")) install.packages("pacman")
pacman::p_load(tm, umap, ggplot2, dplyr, GGally, scales, tidyr)
```

Figure 1: Libraries

```
ds1 <- read.csv("Data/Biomedical_Diabetics_DS1.csv")
ds2 <- read.csv("Data/AI_Diabetics_DS2.csv")
```

Figure 2: Dataset

The code starts by installing the necessary R packages for text mining and visualization and then reads two preprocessed CSV files containing text data about Diabetics biomedicals and AI research papers. ds1 contains biomedical research abstracts and ds2 contains AI-related research abstracts. The libraries tm used for text mining, ggplot2 used for visualization, umap used for clustering, dplyr used for transformation, ggally used for advanced visualization, scales used for Improving axis formatting and data scaling and tidyr used for reshaping and organizing data.

Data Preprocessing:

```r
ds1$domain <- 0
ds2$domain <- 1

merged_data <- rbind(ds1, ds2)
```

Figure 3: Data Preprocessing

For data preprocessing label assigned for ds1(biomedical) domain assigned as 0 and for ds2(AI) domain assigned as 1. rbind used for datasets to be merged for analyses.

Text Cleaning:

```r
corpus <- Corpus(VectorSource(merged_data$Abstract))
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeNumbers)
corpus <- tm_map(corpus, removeWords, stopwords("english"))
corpus <- tm_map(corpus, stripWhitespace)
```

Figure 4: Text Cleaning

Removed noise (punctuation, numbers), removed stopwords and converted to lowercase using corpus in text cleaning.

Save Cleaned Data:

```r
merged_data$Cleaned_Abstract <- sapply(corpus, as.character)
ds <- merged_data[!is.na(merged_data$Cleaned_Abstract) & merged_data$Cleaned_Abstract != "", ]
write.csv(ds, "ids_final_dataset_group_11.csv", row.names = FALSE)
```

Figure 5: Clean Data Save

Here, clean data stored also invalid rows removed and dataset saved for further analysis.

Feature Extraction (TF-IDF):

```r
corpus_final <- Corpus(VectorSource(ds$Cleaned_Abstract))
dtm <- DocumentTermMatrix(corpus_final, control = list(weighting = weightTfIdf))
tfidf_matrix <- as.matrix(dtm)
```

Figure 6: TF-IDF

In figure 6, converted text into numerical form also highlights important words using TF-IDF.

Dimensionality Reduction (UMAP):

```r
set.seed(123)
umap_results <- umap(tfidf_matrix)
umap_df <- as.data.frame(umap_results$layout)
colnames(umap_df) <- c("UMAP1", "UMAP2")
```

Figure 7: UMAP

Reused high-dimensional data to 2D and enable visualization of patterns.

Domain Labeling for Visualization:

```r
umap_df$domain <- ds$domain
umap_df$domain_name <- factor(umap_df$domain, labels = c("Biomedical", "AI"))
```

Figure 8: Domain Labeling for Visualization.

Converted numeric labels into readable names.

Clustering (K-Means):

```r
set.seed(123)
clusters <- kmeans(umap_df[, c("UMAP1", "UMAP2")], centers = 5)
umap_df$Cluster <- as.factor(clusters$cluster)
ds$Cluster <- as.factor(clusters$cluster)
```

Figure 9: K-Means

Divided data into 5 clusters. Each cluster represents a group of similar research topics.

Cluster Analysis:

```r
cluster_table <- table(umap_df$Cluster, umap_df$domain)
cluster_pct <- prop.table(cluster_table, 1) * 100
```

Figure 10: Cluster Analysis

Cluster Analysis shows the percentage of AI vs Biomedical in each cluster.

Cross-Domain Mapping:

```r
mapping_summary <- as.data.frame(cluster_pct)
colnames(mapping_summary) <- c("Cluster", "Domain", "Percentage")
mapping_summary <- pivot_wider(mapping_summary, names_from = Domain, values_from = Percentage)
colnames(mapping_summary) <- c("Cluster", "Bio_Pct", "AI_Pct")
```

Figure 11: Cross-Domain Mapping

Converted cluster data into readable formats.

Topic Classification:

```r
mapping_summary$Topic_Type <- ifelse(mapping_summary$Bio_Pct > 70, "Biomedical Specific",
                                     ifelse(mapping_summary$AI_Pct > 70, "AI Specific",
                                            "Shared / Cross-Domain"))
```

Figure 12. Topic Classification

Clusters are classified into Biomedical-specific, AI-specific and shared.

Keyword Extraction:

```r
get_top_keywords <- function(matrix, clusters, n = 10) {
  top_keywords <- list()
  for (i in unique(clusters)) {
    cluster_matrix <- matrix[clusters == i, , drop = FALSE]
    avg_tfidf <- colMeans(cluster_matrix)
    top_words <- sort(avg_tfidf, decreasing = TRUE)[1:n]
    top_keywords[[as.character(i)]] <- names(top_words)
  }
  return(top_keywords)
}
cluster_keywords <- get_top_keywords(tfidf_matrix, ds$Cluster)
```

Figure 13: Keyword Extraction

Extracted top words from each cluster and helps identify topic meaning.

- ## Visualization

UMAP Cluster Plot:

```r
ggplot(umap_df, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(aes(shape = domain_name), size = 2, alpha = 0.6)
```

Figure 14: UMAP Cluster Plot

Showed clustering and domain distribution.

Domain Distribution:

```r
ggplot(umap_df, aes(x = Cluster, fill = domain_name)) +
  geom_bar(position = "fill") +
```

Figure 15: Domain Distribution

Showed percentage composition.

Boxplot:

```r
ggplot(umap_df, aes(x = Cluster, y = UMAP2, fill = Cluster)) +
  geom_boxplot() +
```

Figure 16: Boxplot

Violin Plot:

```
ggplot(umap_df, aes(x = domain_name, y = UMAP1, fill = domain_name)) +
  geom_violin(trim = FALSE) +
```

Figure 16: Violin Plot

Results & Analysis:

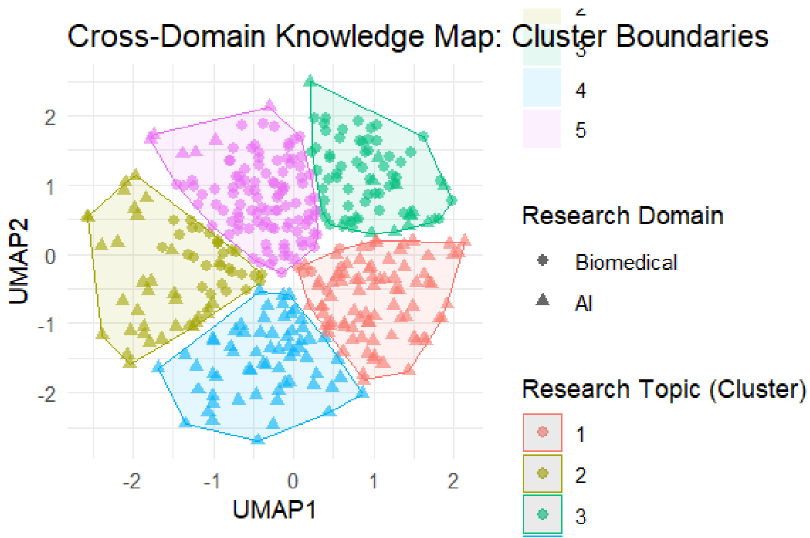

Figure 17: Main Knowledge Mapping

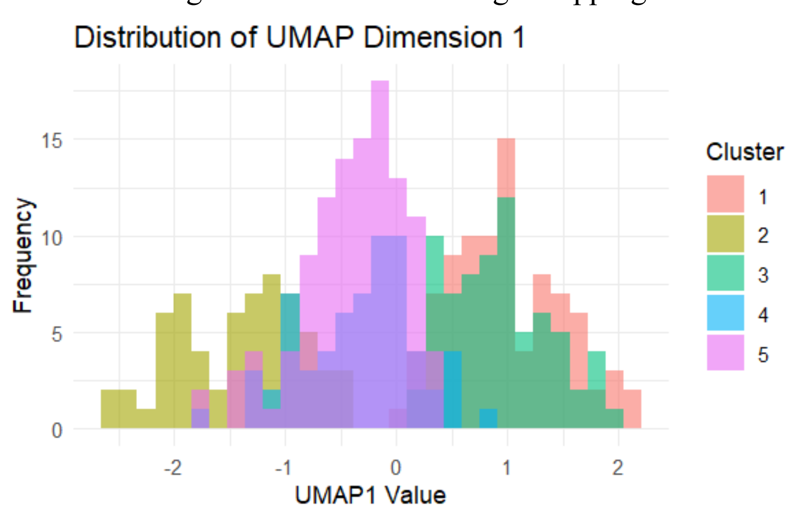

Figure 18: Histogram Plot

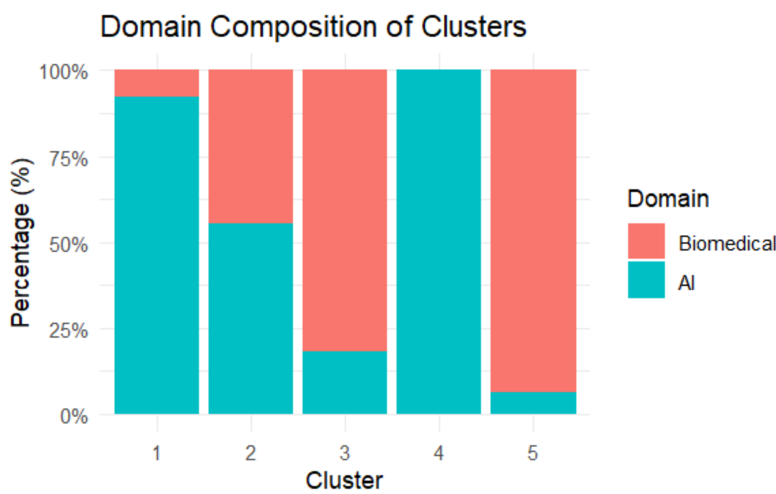

Figure 19: Bar Graph Plot

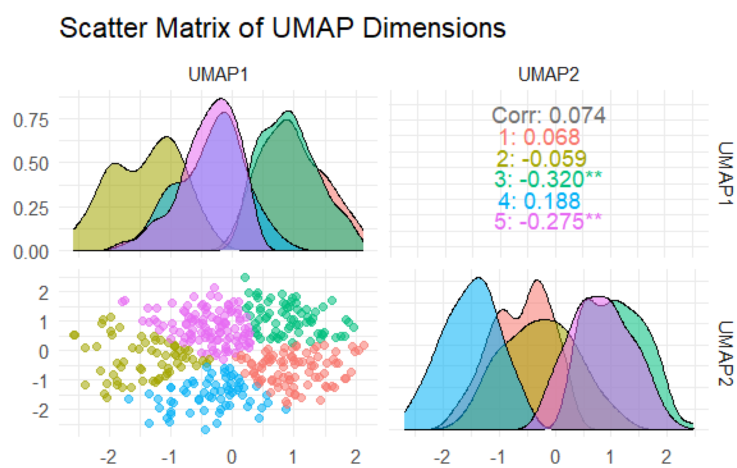

Figure 20: Scatter Matrix Plot

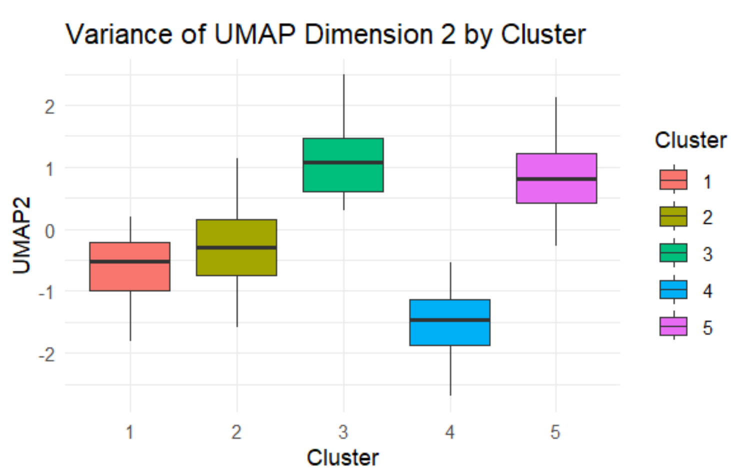

Figure 21: Box Plot

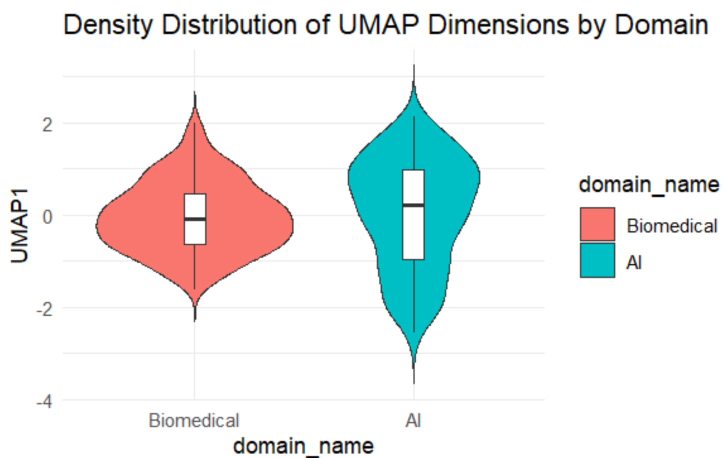

Figure 22: Violin Plot

5 Clusters were identified representing different research topics. Some clusters are purely Biomedical or AI. Some clusters are mixed → cross-domain research. AI domain focuses on Models, algorithms, learning, Biomedical domain focuses on patients, diseases, treatments and overlap shows AI applications in healthcare.

Conclusion

Successfully analyzed and compared biomedical and AI research data using text mining techniques.TF-IDF captured meaningful features, UMAP revealed hidden patterns, Clustering identified domain-specific and shared topics, and the results highlight the growing integration of AI in biomedical research.